

Dhruv H

8884293860 | dhruvh0530@gmail.com | linkedin.com/in/dhruv-trivedi-3005d | github.com/dhruv-0512 | dhruvtrivedi.me

SUMMARY

Information Science undergrad passionate about building robust systems that solve real world problems at scale. Drawn to complex problems where engineering precision and creative thinking come together to build something that lasts.

EDUCATION

B.M.S College of Engineering <i>Bachelor of Engineering in Information Science</i> CGPA: 9.232/10.0	Bengaluru, Karnataka <i>Aug. 2023 – Present</i>
Base PU College <i>Pre-University Course (PUC)</i> Percentage: 92.16%	Bengaluru, Karnataka <i>2021 – 2023</i>
V.V.S Sardar Patel High School <i>Secondary School Leaving Certificate (SSLC)</i> Percentage: 99.36%	Bengaluru, Karnataka <i>2021</i>

PROJECTS

- dvara** | *Python, FastAPI, Redis, Bloom Filter, PyPI* | [visit](#) ↗
- Published a PyPI package for malicious URL detection using a two stage Redis → PostgreSQL Bloom Filter pipeline, achieving 145K URLs/sec throughput.
 - Compressed 268K malicious URLs into 5.14MB at 0.1% FPR; production deployed and publicly available on PyPI, Redis bit checks handle clean traffic entirely, eliminating DB load for benign URLs.
- FluxGate** | *React, FastAPI, Redis, PostgreSQL, Python*
- Built a distributed rate limiting service with 4 algorithms (token bucket, sliding window, leaky bucket) using Redis Lua atomic scripts to eliminate race conditions in concurrent environments.
 - Load tested at 4K+ requests across 20 concurrent clients with per key enforcement; implemented hot config reload and real time WebSocket analytics dashboard showing live RPS and throttle metrics.
- lostoping** | *Python, AIS Data, Isolation Forest, Geospatial*
- Built a multi signal anomaly detection pipeline over 7.3M+ real world data points, scoring 4,621 entities across blackout, loitering, and erratic movement patterns using Isolation Forest.
 - Flagged 34 high risk vessels for sanctions evasion & illegal fishing; outputs interactive maps and HTML intelligence summaries.
- Exoplanet Atmospheric Retrieval** | *Python, XGBoost, Scikit-learn*
- Benchmarked MLP, SVR and XGBoost as lightweight alternatives to Bayesian atmospheric retrieval on the Ariel Big Data Challenge dataset, retrieving 5 gas abundances + equilibrium temperature from transmission spectra.
 - Preprocessing (52 bin wavelength interpolation, log scaling, feature augmentation) proved the dominant performance factor; XGBoost generalized best to real JWST NIRSpec observations of WASP-39b, matching published results.
- SybilGAT** | *Python, PyTorch, Graph Attention Networks*
- Implemented a 2 layer GAT with skip connection & BatchNorm on cresci-2017 (14K users, 203K edges), achieving 97.12% accuracy and 96.18% Macro F1 across 9 bot/human classes.
 - Proved via Mann-Whitney U ($p < 10^{-41}$) that attention weights suppress bot→human camouflage edges; added per account explainability via GNNExplainer.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, HTML/CSS
Frameworks/Libraries: FastAPI, Express.js, Node.js, React, Spring Boot, PyTorch, LangChain, Pandas, NumPy
Infrastructure & Cloud: Redis, PostgreSQL, MySQL, MongoDB, ChromaDB, Docker, Docker Compose, AWS
Developer Tools: Git, GitHub, VS Code, UNIX, Postman, Jupyter Notebook, Claude Code, Cursor
Concepts: Data Structures & Algorithms, Probabilistic Data Structures, Distributed Systems, OOP, REST APIs, Microservices, Machine Learning, Deep Learning, Graph Neural Networks, Retrieval Augmented Generation (RAG), Vector Embeddings, WebSockets, Lua, Competitive Programming (Leetcode/Hackerrank)

CERTIFICATIONS

- Nutanix Certified Associate (NCA)**
- Certified in hyperconverged infrastructure, cloud computing fundamentals, and Nutanix AOS administration.
- JPMorgan Chase: Software Engineering Job Simulation (Forage)**
- Built Kafka integrated Spring Boot microservice with JPA persistence, REST endpoints, and Maven based testing.
- NVIDIA Generative AI Explained**
- Covered fundamentals of generative AI including GANs, transformers, and their practical applications in modern software systems.